Tavishi Session 19

Saturday, July 4, 2026 8:33 PM

Basics
- Variables ✓
- Data Types ✓
- Type Conversion ✓
- Input and Output ✓
- Comments ✓
- Operators

Control Flow
- if
- if-else
- if-elif-else
- Nested Conditions
- Match Case (optional)

Loops
- for Loop
- while Loop
- Nested Loops
- Loop Control Statements (break, continue, pass)

Strings
- String Indexing
- String Slicing
- String Methods
- String Formatting

Collections
- Lists
- Tuples
- Sets
- Dictionaries

Collection Operations
- List Methods
- Tuple Operations
- Set Methods
- Dictionary Methods

Functions
- Function Definition
- Parameters and Arguments
- Return Statement
- Default Arguments
- Keyword Arguments
- Variable Length Arguments (*args, **kwargs)
- Scope (Local and Global)
- Lambda Functions
- map()
- filter()
- reduce()

Variables:- Reserved memory boxes are called variables.

age = 13   name = "Tavishi"

age: 13 ✓   name: Tavishi

Rules of defining variables:-

1. Names can contain letters, digits and underscores (_).
2. The first character cannot be a digit.
3. Names are case-sensitive, so myVar and myvar are treated differently.
4. Keywords such as if, else and for cannot be used as variable names.

1name ✗

_name ✓

tavishi
TAVISHI] → different

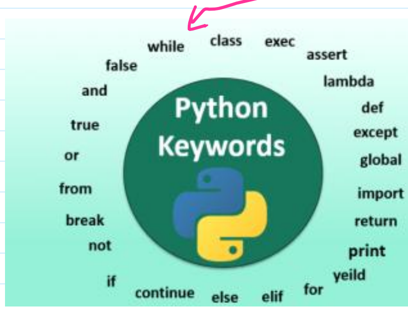
Quiz ;

- student_name ✓
- 2students ✗
- _count ✓
- first-name ✗
- total_marks ✓
- class ✗
- userAge ✓
- user age ✗ spaces not allowed
- price_2026 ✓
- 2026_price ✗
- __init__ ✗ → Exception
- myVariable ✓
- salary$ ✗
- employee_id ✓

spaces not allowed

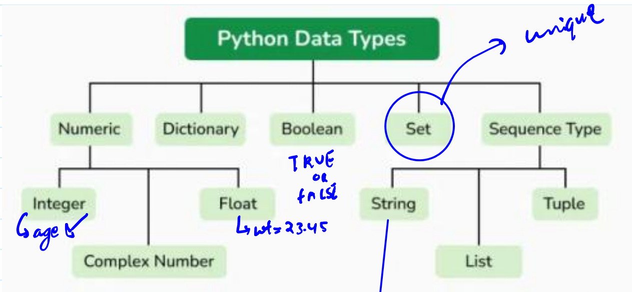
Keywords

while ✗



- 2026_price ✗
- __init__ ✗ → Exception
- myVariable ✓
- salary$ ✗
- employee_id ✓ —
- for ✗
- is_active ✓
- roll_no_1 ✓
- #count ✗
- PI_VALUE ✓
- student@name ✗

Data types:-



unique

TRUE or FALSE

↳ age ✓

↳ wt = 23.45

A B C D E
[10, 20, 19, 18,] = Marks

sequence of character → "Tavishi"

Type Conversion

age = input( ) → By default input is taken as String

print(2 * age)

"7" → "77"

"7" → 7

string   Integer

age = int(input( ))

➢ Manually/Explicitly specifying or changing the data type is called Type conversion

```
age=int(input("Enter the age "))
print(2*age)
```

Comments

↳ It is ignored by python

↳ documentation

Operators

operators → operators

Types of operator:

round off value

10//3 = 3

10/3 = 3.33

1) Arithmetic Operators:-

+ → Addition
- → subtraction
// → floor division
% → modulus ×
** → Exponentiation ×
* → multiplication
/ → Division

Modulus(%):-

↳ It provides remainder

Ex- 10 % 3 = 1

10 % 5 = 0

3)10(3
-9
1

5)10(2
-10
0

Note:- a % b

if a < b
↓
ans = a

↳ 2 % 5 = 2

Exponentiation:- (Power)

3 ** 2 = 9

4 ** 3 = 64

Comments:-

1
2 → ignore by python
3

This is a comment

2) Assignment operator :-

+=
-=
*=
%=

**= ✓
// =

=10 ✓
+=10
print

O/P = 20

$x = x + 10 = 20$

$\hookrightarrow x = x + 10$

(previous value)

$x = 5$

$x += 4$

print (x) O/P = 9

$x -= 1 \rightarrow x = x - 1 = 9$

$x += 4$

$x = x + 4 = 9$

$\boxed{9}$

3) Relation operator :-

==	Equal To ✓
>	Greater Than ✓
<	Lesser Than ✓
>=	Greater Than Equal To ✓
<=	Lesser Than Equal To
!=	Not Equal To

$x=10 \quad y=10$

$x=2 \quad y=10$

$a=10 \quad b=11$

$a>b \rightarrow 10>11$ × false

$x=10 \quad y=2$

$x \geq 2 \rightarrow \underbrace{x>2 \text{ or } x==2}_{\downarrow}$

True

!=

$x=10 \quad y=5$

$x != y \rightarrow$ True

$x=10 \quad y=10$

$x != y \rightarrow$ false

4) Logical operators :-

AND

Both cond^n → should be True

$x = 10$

$y = 5$

$z = 3$

print(x>y and y>z) → True

print(x>y and y<z) → false

OR

Any one need to be true

$x = 10$

$y = 5$

$z = 3$

print(x>y OR y>z) → True

print($\overset{\text{True}}{x>y}$ OR $\overset{\text{False}}{y<z}$) → True